INTERNSHIP PROJECT

Project No: 1) AWS Multi-Region Disaster Recovery Architecture

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1. Introduction

In modern cloud environments, applications must remain available even during system failures, infrastructure outages, or regional disruptions. Disaster Recovery (DR) strategies ensure that applications and data can recover quickly after such failures.

This project demonstrates the implementation of a Multi-Region Disaster Recovery Architecture using AWS. The system replicates resources from the Primary Region (Mumbai) to a Disaster Recovery Region (Singapore) to ensure high availability and fault tolerance.

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2. Objectives

- Design a Multi-Region Cloud Architecture
- Implement Disaster Recovery strategies
- Configure RDS database replication
- Implement S3 Cross-Region Replication
- Test failover scenarios
- Understand RTO and RPO concepts

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3. AWS Services Used

Amazon EC2 – Hosting application server

Amazon RDS – Relational database service

Amazon S3 – Object storage with cross-region replication

IAM – Identity and Access Management

VPC – Virtual private networking environment

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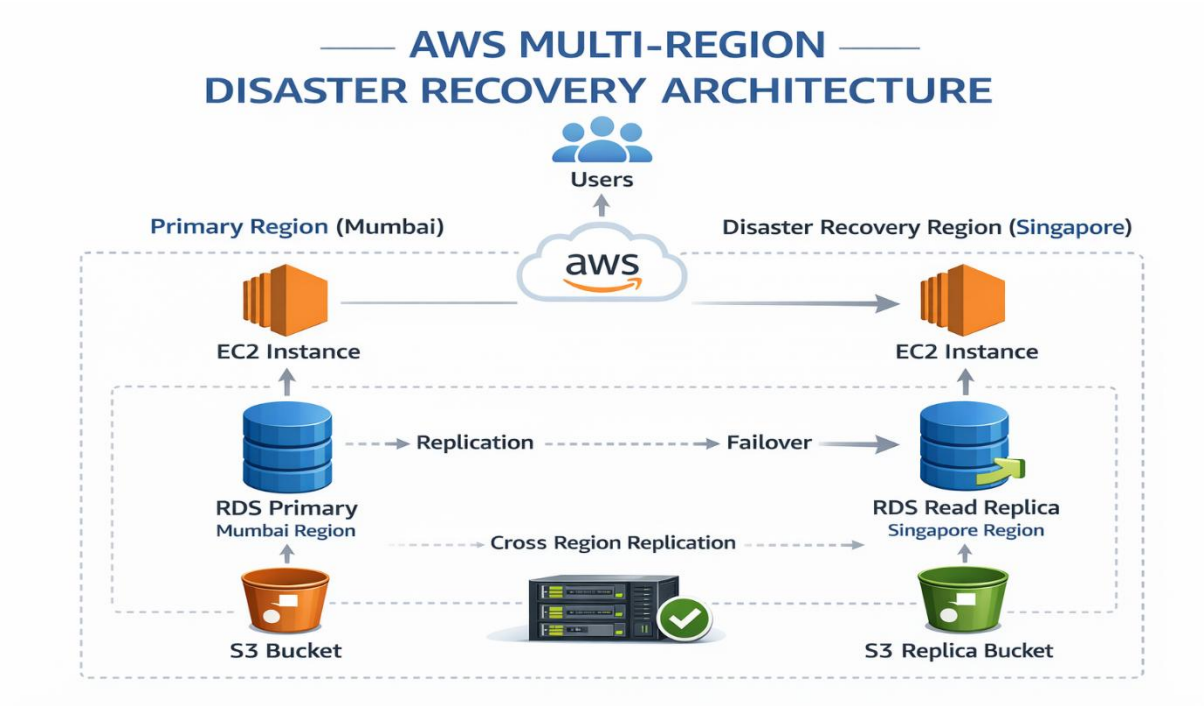
4. Architecture Overview

The architecture consists of two AWS regions.

Primary Region – Mumbai
- EC2 Instance (Application Server)
- RDS Primary Database
- S3 Storage Bucket

Disaster Recovery Region – Singapore
- EC2 Instance (DR Application Server)
- RDS Read Replica
- S3 Replica Bucket



AWS Multi-Region Disaster Recovery Architecture

5. EC2 Instance Setup

An EC2 instance was launched in the Mumbai region to host the web application.

Steps performed:

1. Launch EC2 instance
2. Configure security group
3. Install web server packages

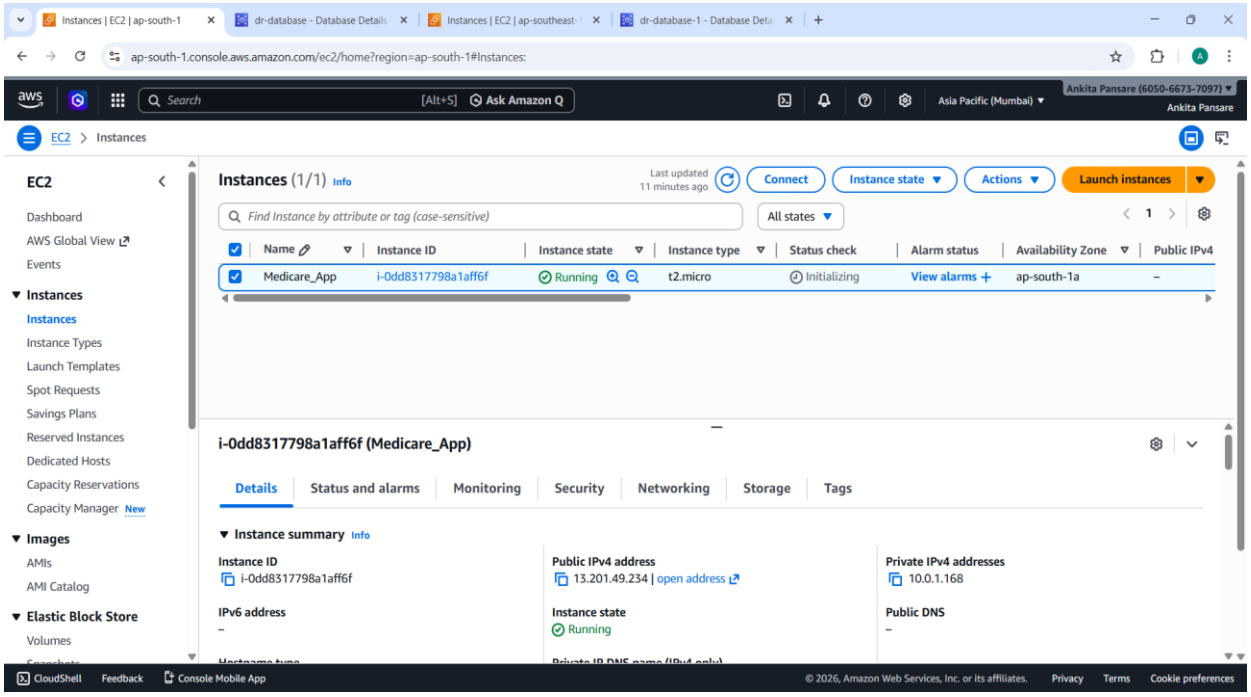
Commands used:

sudo yum update -y

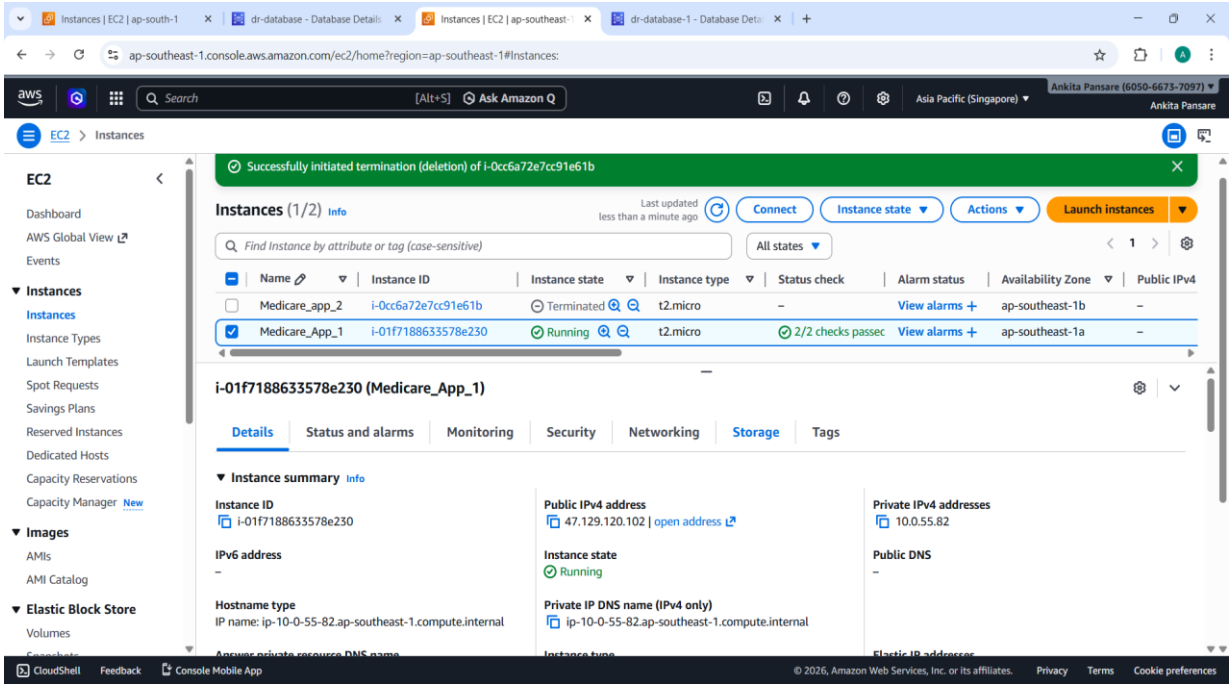
sudo yum install nginx -y

sudo yum install php php-fpm -y

sudo yum install mariadb105-server -y



EC2 Instance created in Mumbai region (Primary_Ec2)



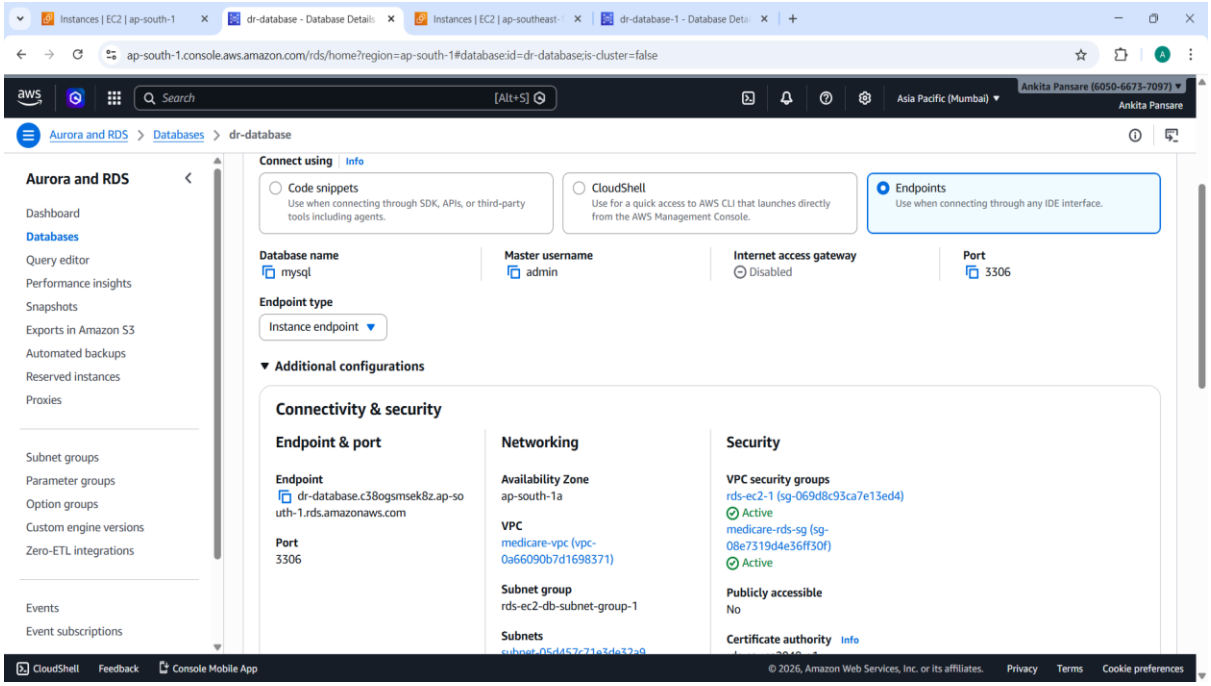
EC2 Instance created in Singapore region (Secondary_Ec2)

6. Database Setup (Amazon RDS)

An Amazon RDS database instance was created to store application data.

Steps performed:

1. Create RDS instance
2. Configure database engine
3. Configure security group
4. Connect application to database



RDS created in Mumbai region (Primary_RDS)

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1. Database connection command:

   mysql -h <your-rds-endpoint> -u <admin> -p

2. Database verification commands:

   SHOW DATABASES;

   USE medicare;

   SHOW TABLES;

   SELECT * FROM appointments;

```
ec2-user@ip-10-0-55-82:~
| Database           |
+--------------------+
| information_schema |
| medicare           |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.009 sec)

MySQL [(none)]> use medicare;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [medicare]> select * from  appointments;
+----+----------------+------+------------+
| id | name           | age  | problem    |
+----+----------------+------+------------+
|  1 | Ankita         |   21 | high fever |
|  2 | Ankita         |   21 | High fever |
|  3 | ankita         |   21 | fever      |
|  4 | Aniket pansare |   18 | cough      |
|  5 | priya sharma   |   30 | Diabetes   |
|  6 | arjune rane    |   40 | asthma     |
+----+----------------+------+------------+
6 rows in set (0.001 sec)

MySQL [medicare]> S
```

Amazon RDS Database

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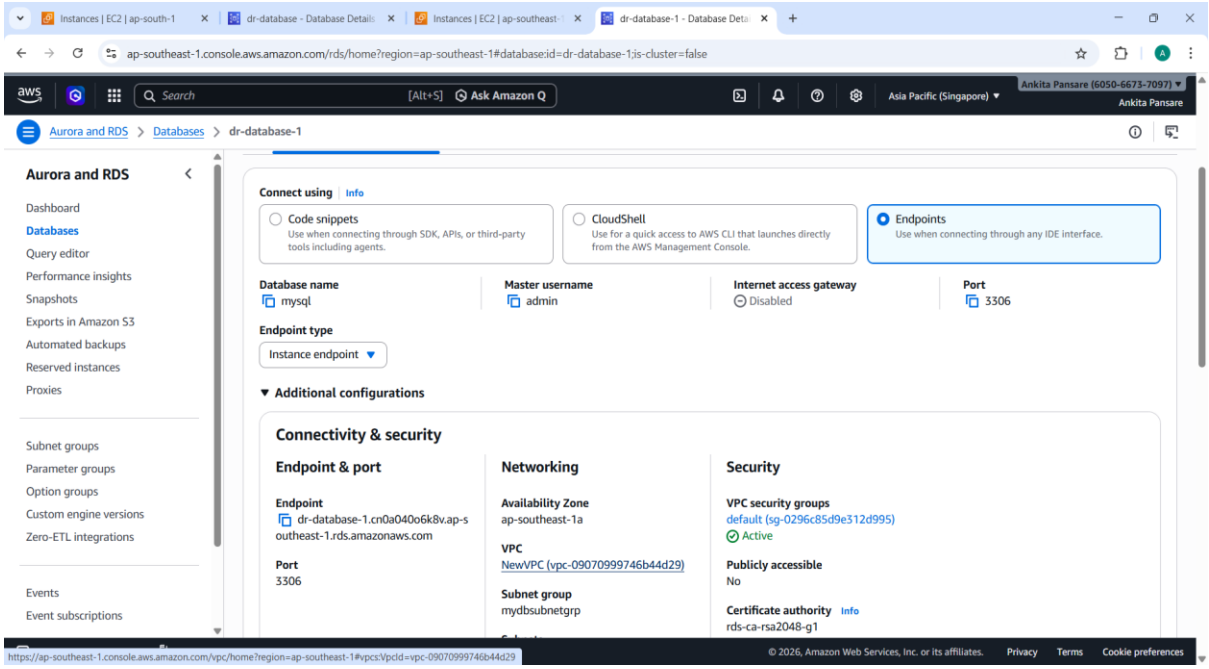
7. RDS Read Replica Configuration

To enable disaster recovery, a Read Replica of the primary RDS database was created in the Singapore region.

Steps performed:

1. Select RDS instance
2. Create Read Replica
3. Choose DR region (Singapore)
4. Verify replication status

During disaster recovery testing, the Read Replica was promoted to a standalone database instance.



RDS Read Replica in Singapore region

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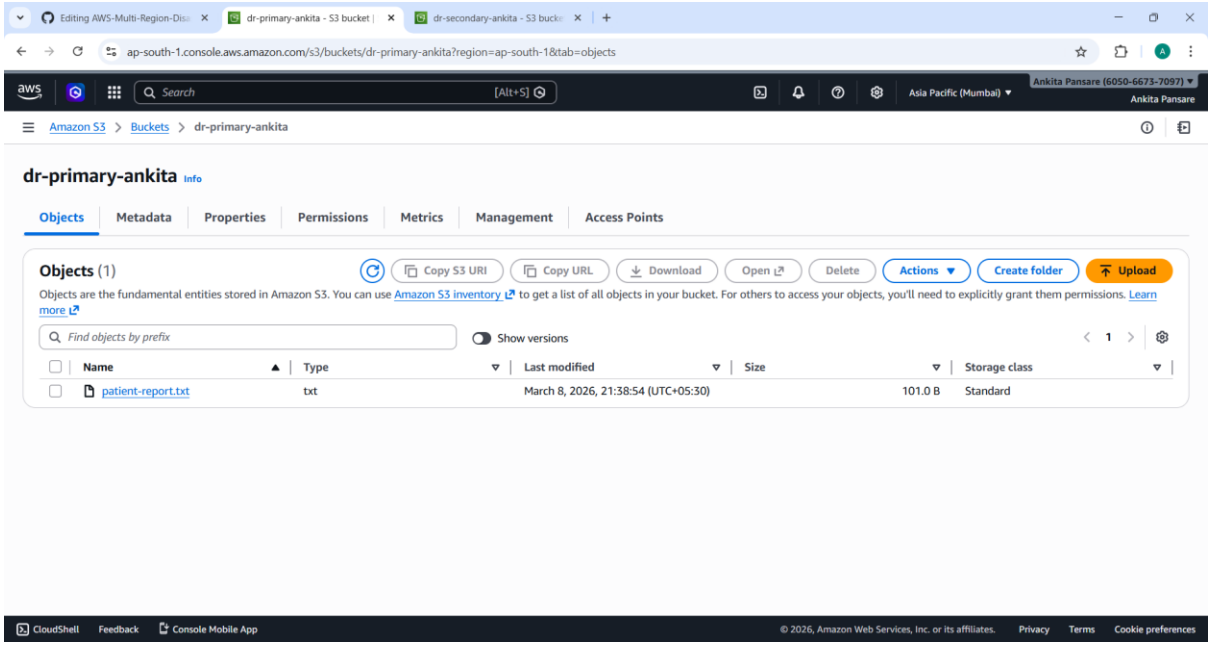
8. S3 Cross-Region Replication

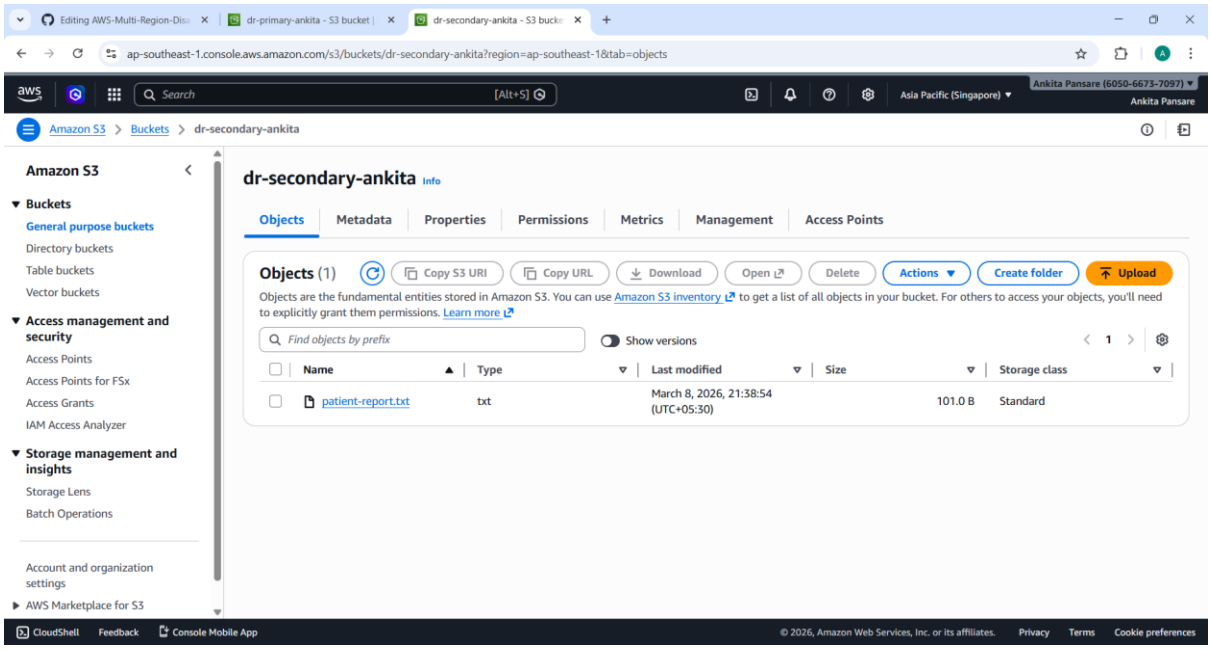
Amazon S3 Cross-Region Replication (CRR) was configured to automatically replicate objects from the primary S3 bucket to the DR bucket.

Example files stored in S3:

- patient-report.pdf
- user-profile.jpg
- dr-test-file.txt



Primary S3 Bucket (Mumbai)



Secondary S3 Bucket (Singapore) Cross Region Replication

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9. Failover Strategy

Failover ensures that the system continues functioning when the primary region fails.

Failover steps:

1. Detect failure in primary region
2. Promote RDS Read Replica
3. Update application database connection
4. Redirect traffic to DR region

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10. Failover Testing

Failover testing was conducted to verify the disaster recovery setup.

Testing steps:

1. Verify replication status
2. Promote RDS read replica
3. Connect application to DR database
4. Confirm application functionality

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11. Recovery Time Objective (RTO)

Recovery Time Objective defines how quickly the system must recover after a failure.

Estimated RTO for this architecture:

5 – 10 minutes

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12. Recovery Point Objective (RPO)

Recovery Point Objective defines the maximum acceptable data loss.

Since database replication is near real-time, the RPO for this system is minimal.

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13. Lessons Learned

During the development of this project, the following concepts were learned:

- Multi-region cloud architecture design
- Disaster recovery planning
- RDS Read Replica configuration
- Cross-region data replication
- Failover strategy implementation
- Understanding RTO and RPO

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14. Conclusion

This project demonstrates how AWS services can be used to implement a reliable Multi-Region Disaster Recovery architecture.

By using Amazon RDS Read Replica and Amazon S3 Cross-Region Replication, the system ensures that application data and services remain available even during regional failures.